UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

Course Name and Code: Data Structures (CS - 201 L)

Project Proposal

Project Title: Intelligent Route Engine

Submitted to: Ms. Muniba Khan

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1. Abstract

Finding optimal routes between locations is a fundamental problem in computer science with applications in navigation systems, networking, and artificial intelligence. The proposed project, **Intelligent Route Engine**, focuses on solving this problem by modeling locations as a graph and applying classical search and path-finding algorithms to compute efficient routes.

The main objective of this project is to design a routing engine that allows comparison between multiple algorithms, including Breadth-First Search (BFS), Depth-First Search (DFS), Uniform Cost Search (UCS), Dijkstra's Algorithm, and A* Search. The system will analyze how each algorithm explores the graph and determines an optimal path. This project provides strong academic value by reinforcing concepts of graph theory, algorithmic efficiency, and heuristic-based search techniques.

2. Background and Justification

Path-finding using graph structures has been extensively studied in computer science. Early approaches such as BFS and DFS focus on traversal, while cost-based algorithms like Dijkstra and UCS aim to find optimal weighted paths. More advanced techniques such as A* enhance efficiency by incorporating heuristics. While these algorithms are usually studied individually, this project integrates them into a single system, enabling structured comparison and deeper understanding. This approach enhances learning by demonstrating real-world relevance of theoretical algorithms.

3. Project Methodology

The proposed system will represent locations as nodes and connections as edges using a graph implemented through an adjacency list. Edge weights will represent distance or cost where required. The user will select a starting location, destination, and search algorithm. Based on this selection, the system will execute the corresponding algorithm to determine the optimal route. Data structures such as queues, stacks, and priority queues will be used according to algorithmic requirements, and the final path along with its cost will be displayed.

4. Project Scope

The Intelligent Route Engine will provide:

- Route calculation between two locations

- Support for multiple search algorithms
- Cost-based and traversal-based routing
- Clear textual output of selected paths

The system will not provide:

- Real-time navigation or GPS integration
- Graphical map visualization
- Large-scale real-world datasets

The scope is limited to an academic environment focused on demonstrating and analyzing algorithm behavior.

5. High Level Project Plan

Activity	Duration	Resources
Problem analysis & graph design	1 week	Team members, DSA notes
Implementation of BFS & DFS	1 week	C++, STL
Implementation of UCS & Dijkstra	1 week	Priority Queue
Implementation of A* & testing	1 week	Heuristic functions
Documentation & submission	1 week	MS Word

6. References

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